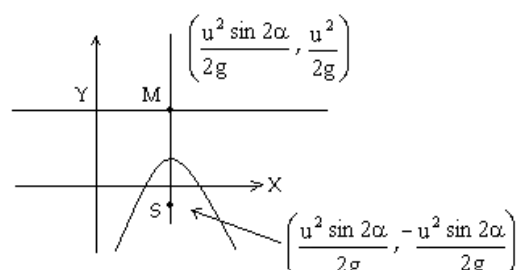


1. (a)
Let any point on it be (x, y) then by definition,
$$\sqrt{(x+1)^2 + (y-1)^2} = \frac{1}{2} \left| \frac{x-y+3}{\sqrt{1^2+1^2}} \right|$$

Squaring and simplifying, we get
 $7x^2 + 2xy + 7y^2 + 10x - 10y + 7 = 0$, which is the required ellipse
2. (c)
The locus of point of intersection of two perpendicular tangents drawn on the ellipse is $x^2 + y^2 = a^2 + b^2$, which is called "director circle".
Given ellipse is $\frac{x^2}{9} + \frac{y^2}{4} = 1$.
 $\therefore$ Locus is $x^2 + y^2 = 9 + 4$, i.e. $x^2 + y^2 = 13$
3. (b)
Let the equations of the circle and that of the parabola be $x^2 + y^2 + 2gx + 2fy + c = 0$ and $y^2 = 4ax$ respectively.
Eliminating x from these equation, we get
 $y^4 + y^2(16a^2 + 8ag) + 32a^2fy + 16a^2c = 0 \Rightarrow y_1 + y_2 + y_3 + y_4 = 0$.
4. (b)
The slopes of the tangents to the parabola $y^2 = 4ax$ passing through (x_1, y_1) are given by the equation $m^2x_1 - my_1 + a = 0$
Given $(x_1, y_1) = (1, 4)$, $a = 3$.
 $\therefore$ Slopes are given by $m^2 \cdot 1 - m \cdot 4 + 3 = 0$
 $m^2 - 4m + 3 = 0$
 $\Rightarrow m = 1, 3$.
5. (c)
We know that from $(c, 0)$ three distinct normals can be drawn to a parabola if $c > 2a$
 $\frac{1}{4}$
i.e., $c > 2 \cdot \frac{1}{4}$
 $\frac{1}{2}$
i.e., $c > \frac{1}{2}$
6. (a)
 $\Delta \neq 0$, $h^2 = ab$ i.e., parabola
7. (d)
According to the figure, the length of latus rectum is $2(SM) =$
$$\frac{u^2}{2g} (1 + \cos 2\alpha) = \frac{2u^2 \cos^2 \alpha}{g}$$

 $2 \times \frac{u^2}{2g}$

8. (d)
Let P $(a \sec \theta, b \tan \theta)$ be any point on the hyperbola
 $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Equation of the chord of contact of tangents from P to the hyperbola
 $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 2$ is
 $\frac{x a \sec \theta}{a^2} - \frac{y b \tan \theta}{b^2} = 2$
 $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 2$ (1)
The two hyperbolas have a common set of asymptotes $y = \pm \frac{b}{a} x$.
 $\frac{b}{a} x$ meets the chord of contact of tangents at Q
 $\left(\frac{2a}{\sec \theta - \tan \theta}, \frac{2b}{\sec \theta - \tan \theta} \right)$
 $-\frac{b}{a} x$ meets the chord of contact at
 $R \left(\frac{2a}{\sec \theta + \tan \theta}, \frac{-2b}{\sec \theta + \tan \theta} \right)$
Area of ΔOQR
$$= \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ \frac{2a}{\sec \theta - \tan \theta} & \frac{2b}{\sec \theta - \tan \theta} & 1 \\ \frac{2a}{\sec \theta + \tan \theta} & \frac{-2b}{\sec \theta + \tan \theta} & 1 \end{vmatrix}$$

 $= 4ab \text{ sq. units}$
9. (b)
Equation of the chord connecting the points $(a \sec \theta, b \tan \theta)$ and $(a \sec \phi, b \tan \phi)$ is
 $\frac{x}{a} \cos \left(\frac{\theta - \phi}{2} \right) - \frac{y}{b} \sin \left(\frac{\theta + \phi}{2} \right) = \cos \left(\frac{\theta + \phi}{2} \right)$ If it passes through $(ae, 0)$, we have, $e \cos \left(\frac{\theta - \phi}{2} \right) = \cos \left(\frac{\theta + \phi}{2} \right)$
$$\Rightarrow e = \frac{\cos \left(\frac{\theta + \phi}{2} \right)}{\cos \left(\frac{\theta - \phi}{2} \right)} = \frac{1 - \tan \frac{\theta}{2} \cdot \tan \frac{\phi}{2}}{1 + \tan \frac{\theta}{2} \cdot \tan \frac{\phi}{2}}$$

$$\Rightarrow \tan \frac{\theta}{2} \cdot \tan \frac{\phi}{2} = \frac{1 - e}{1 + e}$$
10. (c)

$(3x + 4y - 5)(4x + 3y - 5) = \lambda$ is family of hyperbola have given asymptotes as it passes through origin $\lambda = 25$.

$$12x^2 + 12y^2 + 25xy - 35x - 35y = 0.$$

11. (a)

If $lx + my + n = 0$ is a tangent to the parabola $x^2 = 4ay$, then $al^2 = mn$.

$\therefore$ If $x + y + k$ is a tangent to the parabola $x^2 = 4y$ then $1 \cdot 1^2 = 1 \cdot k \Rightarrow k = 1$.

12. (c)

$y' = 2(x - 1) = 1$ gives the point $\left(\frac{7}{2}, \frac{1}{4}\right)$ and the required

tangent is $y - \frac{1}{4} = 1 \cdot \left(x - \frac{7}{2}\right)$ or
 $4y - 4x + 13 = 0$.

13. (a)

Equation of line TQ: $y = at$

Find Q point, Q: $\left(\frac{at^2}{4}, at\right)$ N: $(at^2, 0)$

To find T point solve the line NQ with y-axis.

14. (d)

$$y^2 = ax \Rightarrow 2y \cdot \frac{dy}{dx} = a$$

$$\text{Slope of tangent } \frac{dy}{dx} = \frac{a}{2y} \quad \dots (i)$$

$$\text{Slope of given tangent } m = \tan 45^\circ = 1 \quad \dots (ii)$$

$$\text{From (i) and (ii) are same } \therefore \frac{a}{2y} = 1 \Rightarrow y = \frac{a}{2}$$

$$\text{and } x = a/4 \Rightarrow (a/4, a/2)$$

15. (b)

$$\frac{x^2}{16} - \frac{y^2}{9} = 1$$

$y = 3/4 x$ [equation of asymptotes]

line $\perp$ to asymptotes and passing through centre (diameter)

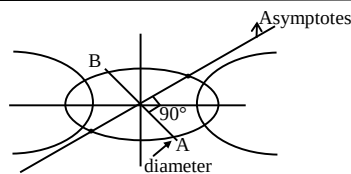
$$y = -\frac{4}{3}x$$

point of intersection of line (diameter)

$$y = -\frac{4}{3}x \text{ and ellipse } \frac{x^2}{16} - \frac{y^2}{9} = 1$$

$$A = \left(\frac{45}{\sqrt{481}}, -\frac{60}{\sqrt{481}}\right) B = \left(-\frac{45}{\sqrt{481}}, \frac{60}{\sqrt{481}}\right)$$

$$\text{Diameter} = \sqrt{\left(\frac{90}{\sqrt{481}}\right)^2 + \left(\frac{120}{\sqrt{481}}\right)^2} = \frac{150}{\sqrt{481}}$$



16. (b)

$$x \cos \alpha + y \sin \alpha = p$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

intersection point A and B

join eqⁿ of OA, OB

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = \left(\frac{x \cos \alpha + y \sin \alpha}{p}\right)^2$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{x^2 \cos^2 \alpha}{p^2} - \frac{y^2 \sin^2 \alpha}{p^2}$$

$$\frac{2xyp \cos \alpha \sin \alpha}{p^2} = 0$$

Chord subtend a right angle at the centre.

Coefficient of x^2 + coefficient of y^2 = 0

$$\frac{1}{a^2} - \frac{\cos^2 \alpha}{p^2} - \frac{1}{b^2} - \frac{\sin^2 \alpha}{p^2} = 0$$

$$\frac{1}{a^2} - \frac{1}{b^2} = \frac{1}{p^2}$$

$$\frac{b^2 - a^2}{a^2 b^2} = \frac{1}{p^2} \Rightarrow p = \frac{ab}{\sqrt{b^2 - a^2}}$$

$x \cos \alpha + y \sin \alpha = p$ touches $x^2 + y^2 = r^2$ if length of $\perp$ from $(0, 0)$ upon the line = Radius

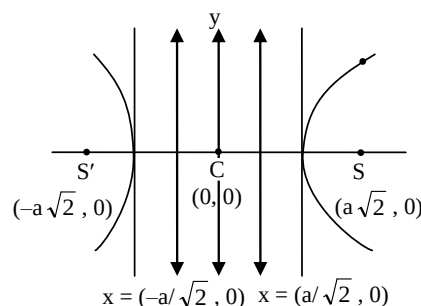
$$\text{if } \frac{p}{\sqrt{\cos^2 \alpha + \sin^2 \alpha}} = r \Rightarrow p = r$$

line touches a fixed circle with radius $\frac{ab}{\sqrt{b^2 - a^2}}$

17. (b)

Let the coordinates of P be (x, y)

The eccentricity of the hyperbola is $\sqrt{1 + \frac{a^2}{b^2}} = \sqrt{2}$



So the coordinates of the foci are $S(a\sqrt{2}, 0)$ and $S'(-a\sqrt{2}, 0)$.

Equation of the corresponding directrices are $x = a/\sqrt{2}$ and $x = -a/\sqrt{2}$.

By definition of the hyperbola

$$SP = e \text{ (distance of P from } x = a/\sqrt{2} \text{)}$$

$$= \sqrt{2} |x - a/\sqrt{2}|$$

$$\text{Similarly } S'P = \sqrt{2} |x + a/\sqrt{2}|$$

$$\text{So that } SP \cdot S'P = 2 |x^2 - a^2/2| = 2x^2 - a^2 = x^2 + y^2 = (CP)^2$$

($\because$ P lies on the hyperbola $x^2 - y^2 = a^2$).

18. (b)

Given equation is $x^2 - 2y^2 - 2 = 0$, it can be rewritten as

$$\frac{x^2}{2} - \frac{y^2}{1} = 1$$

$$\text{Here } a^2 = 2, b^2 = 1$$

We know that equation of hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then the product of length of perpendicular drawn from any point on the hyperbola to the asymptotes is

$$\frac{a^2 b^2}{a^2 + b^2} = \frac{2(1)}{2+1} = \frac{2}{3}$$

19. (b)

Let common tangent is $y = mx + c$... (1)

$\because$ (1) touch $x^2/9 - y^2/16 = 1$

$$\Rightarrow C = \pm \sqrt{9m^2 - 16}$$

$$\because (1) \text{ also touch } \frac{x^2}{(-16)} - \frac{y^2}{(-9)} = 1$$

$$\Rightarrow c = \pm \sqrt{-16m^2 + 9} \Rightarrow 9m^2 - 16 = -16m^2 + 9$$

$$\Rightarrow 25m^2 = 25 \Rightarrow m^2 = 1 \Rightarrow m = \pm 1$$

20. (b)

Let the hyperbola be $xy = c^2$

Tangent at any point $(ct, c/t)$

$$x \frac{dy}{dx} + y = 0 \Rightarrow \left(\frac{dy}{dx} \right)_{(ct, c/t)} = - \frac{c/t}{ct} = - \frac{1}{t^2}$$

$$\text{Equation of tangent } \Rightarrow y - \frac{c}{t} = \frac{-1}{t^2} (x - ct)$$

$$a_1 \text{ (Intercept on x-axis)} ; 0 - \frac{c}{t} = \frac{-1}{t^2} (x - ct)$$

$$a_1 = 2ct ; b_1 \text{ (Intercept of y-axis)} ; b_1 = 2c/t$$

$$\text{Equation of Normal } \Rightarrow y - \frac{c}{t} = t^2 (x - ct)$$

$$a_2 = \frac{c(t^4 - 1)}{t^2}, b_2 = \frac{-c(t^4 - 1)}{t} \Rightarrow a_1 a_2 + b_1 b_2 = 0$$